

RevoFin : Loan Portfolio Analysis



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Loan Portfolio Analysis

Risk Management & Executive Report

Cohort Period : 2012-2019

Target Audience : Risk Management Team
& Executive Management

Portfolio OS

\$2.80B

Total Customers

176,087

TKB30

98.11%

Bad Cohort TKB30

96.42%

Good Cohort TKB30

98.75%



Focus Analysis

Business Model:

Fintech Lending - consumer & SME loans, streamlining the application process, automated credit assessment, offering competitive interest rates

Prioritas:

- Mengukur OS, ENR, TKB30 Portofolio
- Mengidentifikasi anomaly cohort (bad vs good)
- Profil risiko nasabah antar cohort
- Rekomendasi perbaikan underwriting

Problem

Bagaimana RevoFin dapat menstabilkan performa portofolio pinjaman akibat rendahnya tingkat keberhasilan bayar (TKB30) pada cohort 2012 & 2014 yang hanya mencapai ~96.25% & ~96.58% di bawah portfolio average 97.43%, guna mencapai target peningkatan TKB30 sebesar +2–3%, serta memahami profil risikonya berdasarkan karakteristik loan dan nasabah guna mencapai peningkatan efisiensi underwriting (+20%) bagi RevoFin di periode berikutnya?

Objectives

Untuk mengetahui strategi bisnis RevoFin yang dapat meningkatkan TKB30 sebesar 2–3% dengan menstabilkan struktur pinjaman dari tekanan interest rate tinggi dan tenor panjang guna mendorong TKB30 portofolio di atas **97.5%**, mengoptimalkan kebijakan underwriting yang sudah ada, serta meningkatkan kualitas seleksi borrower (+20% efisiensi) guna mencapai target peningkatan performa portofolio pada periode lending berikutnya.

Top Analytical Findings

Primary:

- **Interest Rate Gap** : Bad 19.6% vs Good Cohort 15.08% (Gap:+4.5%) - bunga tinggi langsung meningkatkan beban cicilan borrower sehingga TKB30 Bad Cohort turun signifikan.
- **Loan Tenor Mismatch** : Bad 43.6% vs Good Cohort 32.2% berjangka 60 bulan - kombinasi high interest rate + long tenor menciptakan total repayment burden yang besar -> borrower fatigue -> default risk naik

Secondary:

- **Income Gap** : Bad ~\$75K vs Good Cohort ~\$82–83K (selisih ~10%) menunjukkan kapasitas repayment yang lebih lemah, meski tidak cukup besar untuk menjadi penyebab tunggal.
- **Debt Consolidation Risk** : Bad 59.85% vs Good Cohort 53.65% - borrower yang sudah memiliki beban utang sebelumnya → financial stress lebih tinggi → risiko default meningkat
- **Loan Type Gap** : JOINT loan Good 13.76% vs Bad cohort hanya 5.61%. Joint = dual income → repayment capacity lebih kuat → default risk lebih rendah

Context:

- **Home Ownership & Employment Length** : Distribusi hampur identik antar cohort -> bukan faktor pembeda, fokus screening ke rate, tenor, dan income

Strategic Recommendation

Dynamic Interest Rate Cap: Terapkan rate cap berbasis credit scoring, pisahkan segmen risiko dengan pricing tier ketat agar high-risk borrower tidak otomatis diberi rate tinggi tanpa mitigasi lain.

Tenor Restriction by Credit Profile: Batasi akses tenor 60 bulan hanya untuk borrower dengan credit profile kuat, tambahkan syarat income minimum untuk jangka panjang.

Income-to-Installment Ratio Threshold: Perkuat verifikasi income di underwriting, gunakan income-to-installment ratio sebagai threshold approval, pertimbangkan annual_inc_joint sebagai faktor positif.

Debt Consolidation Screening Layer: Tambahkan layer screening khusus, analisis existing debt, debt-to-income ratio, dan payment history sebelum approval, pertimbangkan rate premium untuk segmen ini.

Insentif Joint Loan Application: Insentifkan joint applications lewat rate discount atau limit yang lebih tinggi, prioritaskan joint dalam approval jika borderline.

Growth-Adjusted Underwriting Standard: Saat volume ekspansi tumbuh cepat, perkuat standard approval, bukan longgarkan, standar approval; lakukan cohort monitoring per kuartal untuk deteksi dini penurunan kualitas.

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Business Overview & Background



RevoFin adalah perusahaan **(fiktif) Fintech Lending (Peminjaman)**. RevoFin menghubungkan peminjam (individu & UKM) dengan produk pinjaman secara online. Profit yang mereka hasilkan dari biaya transaksi (*transaction fees*), pendapatan bunga dari pinjaman (*interest income*), dan kemitraan strategis dengan bisnis yang ingin menyediakan produk keuangan kepada pelanggan mereka. **Risiko utama yang perusahaan ini biasanya hadapi adalah risiko gagal bayar (*default*)**. Jika banyak peminjam tidak membayar kembali, pendapatan bunga hilang dan modal perusahaan tergerus. Dalam dunia lending, ada dua kutub yang harus dijaga keseimbangannya:

- **Growth (Pertumbuhan):** Meminjamkan uang sebanyak-banyaknya (Total Outstanding/OS tinggi).
- **Risk (Risiko):** Memastikan uang itu kembali (TKB30 tinggi).

DARCI Model

D – Decision Maker	Chief Risk Officer (CRO)
A – Accountable	Risk Management Team, Credit Policy Manager
R – Responsible	Data Analyst
C – Consulted	Credit Analyst Team, Actuarial & Data Science, Legal & Compliance
I – Informed	Executive Management, Finance Team, Product & Tech Team





01

Project Introduction

RevoFin : Business Background & Problem

Project Introduction

Company Profile & Backgrounds/Context

Company Profile

- Fintech Lending Platform -> Consumer & Small Business Loans
- Revenue dari Transaction Fees, Interest Income, dan Partnerships
- Online Platform dengan Automated Credit Assessment
- Posisi Data Analyst : Mendukung Keputusan Bisnis Berbasis Data

Business Background

- Portfolio aktif \$2.80B dari 176,087 nasabah
- Loan cohort dari 2012 hingga 2019
- Metrik utama: TKB30 sebagai regulatory metric OJK
- Kebutuhan: monitoring kualitas portofolio secara berkala
- Kebutuhan: monitoring kualitas portofolio secara berkala

Target Audience

- Risk Management Team : Evaluasi kualitas kredit
- Executive Management : strategic decision making
- Output : Presentations deck + dashboard
- Deliverable: Insight & actionable recommendations (OBIPR Framework)

Project Introduction

Company Profile & Backgrounds/Context

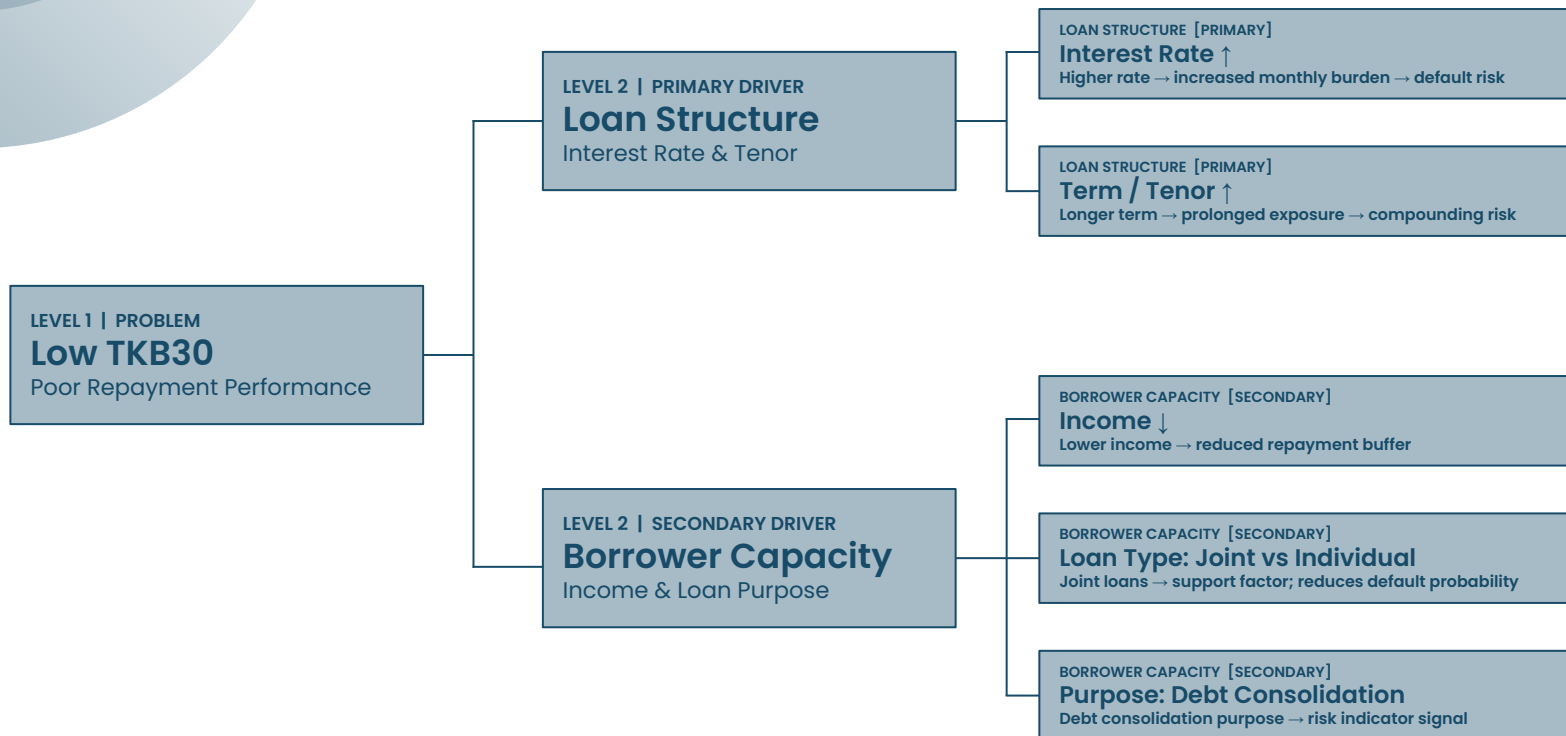
Business Problem

- Apakah portofolio dalam kondisi sehat berdasarkan TKB30?
- Bagaimana mengidentifikasi cohort pinjaman dengan TKB30 tertinggi dan terendah?
- Bagaimana profil risiko nasabah di masing-masing cohort (Good vs Bad Cohort/Anomaly)?
- Apa upaya yang bisa dilakukan untuk menstabilkan performa portofolio pinjaman yang TKB30 nya rendah?
- Apa yang bisa diubah dari proses underwriting untuk mencegah ini?

Objectives

- Specific : Analisis OS, ENR, TKB30 per level (tahun) dan cohort
- Measurable : Kuantifikasi selisih TKB30 antar cohort
- Achievable : SQL Bigquery dari enam tabel dan 270K baris data (records)
- Relevant: Regulatory metric OJK, langsung actionable
- Time-bound: Data historis 2012–2019

Root Cause Analysis (RCA) – Issue Tree



Problem Statement

Bagaimana RevoFin dapat menstabilkan performa portofolio pinjaman akibat rendahnya tingkat keberhasilan bayar (TKB30) pada cohort 2012 & 2014 yang hanya mencapai ~96.25% & ~96.58% di bawah portfolio average 97.43%, guna mencapai target peningkatan TKB30 sebesar +2–3%, serta memahami profil risikonya berdasarkan karakteristik loan dan nasabah guna mencapai peningkatan efisiensi underwriting (+20%) bagi RevoFin di periode berikutnya?

Objective

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02

Data Cleaning

Data Cleaning - SQL BigQuery

Data Cleaning

Standardisasi dataset sebelum analisis

01

Clean customer_id — Tabel Loan & Customer

```
ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id,"[a-zA-Z0-9]"), "")
```

Hapus binary encoding, pertahankan hanya huruf & angka.
Wajib agar JOIN antar tabel berhasil.

02

Filter Active Portfolio — Loan Status

```
WHERE loan_status IN ('Current','In Grace Period',  
'Late (16-30 days)','Late (31-120 days)','Default')
```

Buang Fully Paid (76,361) dan Charged Off (17,851).
Hanya loan aktif yang masuk analisis.

03

Bonus Cleaning — Standardisasi Kolom

```
term:'36 months'→36 | emp_length:'5 years'→5  
home_ownership:ANY/NONE→OTHER | type:JOINT APP→JOINT
```

Konversi format STRING ke numerik dan standardisasi kategori untuk kemudahan analisis.

HASIL CLEANING

Raw Records

270,299

Total semua loan

Active Portfolio

176,087

Setelah filter status

Excluded Loans

94,212

Fully Paid + Charged Off

Tables Joined

2 Tables

loan + customer via clean ID

Cohort Periode

2012–2019

8 tahun data historis



35% data di-exclude
dari 270K → 176K active loans siap analisis

Data Cleaning

1. Clean customer_id

Bersihkan karakter non-alphanumeric (ekstrak hanya huruf dan angka) dari kolom customer_id di tabel customer dan loan:

```
ARRAY_TO_STRING(
  REGEXP_EXTRACT_ALL(customer_id, "[a-zA-Z0-9]"), ""
) AS cleaned_customer_id
```

3. Convert emp_length → Numeric

Ubah "1 year", "5 years", dll menjadi angka integer agar bisa dihitung rata-rata:

```
CAST(
  REGEXP_EXTRACT(emp_length, r'\d+')
  AS INT64
) AS emp_length_num
```

5. Clean home_ownership

Gabungkan kategori ambiguous (ANY, NONE, OTHER) menjadi satu kategori 'OTHER' yang lebih clean:

```
CASE
  WHEN home_ownership
    IN ('ANY', 'NONE', 'OTHER')
    THEN 'OTHER'
  ELSE home_ownership
  END AS home_ownership_clean
```

2. Filter Active Loan Portfolio

Loan aktif yang masuk analisis (kecuali Charged Off & Fully Paid)

```
WHERE loan_status IN ('Current', 'In Grace Period', 'Late
  (16-30 days)', 'Late (31-120 days)', 'Default')
```

4. Clean type (Loan Type)

Samakan penulisan: Individual, INDIVIDUAL → INDIVIDUAL;
Joint App, JOINT → JOINT:

```
CASE
  WHEN UPPER(type)
    IN ('JOINT', 'JOINT APP')
    THEN 'JOINT'
  WHEN UPPER(type) = 'INDIVIDUAL'
    THEN 'INDIVIDUAL'
  ELSE 'OTHER'
  END AS type_clean
```

6. Convert term → Numeric

Ubah "36 months", "60 months" menjadi angka integer 36 dan 60:

```
CAST(
  REGEXP_EXTRACT(term, r'\d+')
  AS INT64
) AS term_num
```



03

Executive Summary

Data Cleaning - SQL BigQuery

Executive Summary

Portfolio-level metrics: OS, ENR, dan TKB30

Total Customers 176,087 Active Unique Borrowers (loan status exclude 'Charged Off' & 'Fully Paid')	Total Customers 176,087 Active Unique Borrowers (loan status exclude 'Charged Off' & 'Fully Paid')	Total Customers 176,087 Active Unique Borrowers (loan status exclude 'Charged Off' & 'Fully Paid')	Total Customers 176,087 Active Unique Borrowers (loan status exclude 'Charged Off' & 'Fully Paid')	Total Customers 176,087 Active Unique Borrowers (loan status exclude 'Charged Off' & 'Fully Paid')
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$$\text{TKB30} = 1 - (\text{OS DPD} > 30 / \text{Portfolio OS}) = 1 - (\$53,042,275 / \$2,801,078,850) = 98.11\%$$
$$\text{ENR} = \text{Portfolio OS} - \text{Default OS} = \$2,801,078,850 - \$195,275 = \$2,800,883,575$$

Loan Status	Total Loan	Total OS	% Portfolio	Kategori DPD	Masuk ENR?
Current	170,461	\$2,715M	96.81%	DPD = 0 (tidak ada keterlambatan)	Ya
In Grace Period	1,638	\$27.6M	0.93%	DPD 1–15 hari (toleransi awal)	Ya
Late (16–30 days)	802	\$13.0M	0.46%	DPD 16–30 hari (terlambat, belum kritis)	Ya
Late (31–120 days)	3,174	\$52.8M	1.80%	DPD 31–120 hari (kritis, kurangi TKB30)	Ya
Default	12	\$195K	0.007%	DPD > 120 hari (gagal bayar)	Tidak



Exploratory Data Analysis

EDA Highlights

Distribusi karakteristik kunci nasabah & pinjaman

TKB30 per Cohort

- Bad Cohort (2012, 2014): 96.25% – 96.58%
- Good Cohort (2018, 2019): 98.67% – 98.83%
- Selisih ~2% — signifikan di dunia credit risk
- TKB30 meningkat dari 96% ke ~99% di cohort terbaru

Borrower Capacity (SECONDARY)

- Annual Income: Bad ~\$75K vs Good ~\$82–83K (~10%)
- Joint Loan: Bad 5.61% vs Good 13.76%
- Debt Consolidation: Bad 59.85% vs Good 53.65%
- Berperan secondary di default risk

Loan Structure (PRIMARY DRIVER)

- Interest Rate: Bad 19.60% vs Good 15.08% (+4.5pp)
- Term 60 bulan: Bad 43.59% vs Good 32.19%
- Deadly combo: high interest + long tenor
- Primary driver utama penurunan TKB30

Non-Driver Variables (CONTEXT)

- Home Ownership: hampir identik antar cohort
- Employment Length: counterintuitive (Bad lebih senior)
- Portfolio Growth >300x: bukan penyebab langsung
- Tidak signifikan untuk membedakan TKB30

Cohort Analysis

Total Customers, OS, ENR, TKB30 per cohort tahunan 2012-2019

Cohort	Total Customers	Total Loans	Portfolio OS	ENR	TKB30	
2012	112	112	\$2.35M	\$2.35M	96.25%	Lowest
2013	1,169	1,169	\$23.99M	\$23.95M	96.90%	
2014	8,215	8,215	\$128.95M	\$128.93M	96.58%	
2015	4,208	4,208	\$84.82M	\$84.82M	97.40%	
2016	37,340	37,340	\$549.21M	\$549.07M	97.61%	Highest
2017	32,177	32,177	\$486.74M	\$486.74M	97.23%	
2018	45,675	45,675	\$738.17M	\$738.16M	98.67%	
2019	47,191	47,191	\$786.86M	\$786.86M	98.83%	

Portfolio Average TKB30: 97.43% | Highlighted = Bad Cohort (2012+2014), Good Cohort (2018+2019) | Tren: TKB30 meningkat konsisten 2012→2019

Executive Summary

SQL Bigquery Code Documentation

STEP 5

Calculate OS, ENR, and TKB30 of each loan cohort. You can define cohorts on a monthly or yearly basis

STEP 6

From your findings in (5), is there any particular cohort with high or low TKB30?

Result Query:

Row	cohort_year	total_customers	os	enr	tkb30
1	2012.0	112	2346175.0	2346175.0	0.962481485822...
2	2013.0	1169	23986075.0	23951075.0	0.968954904043...
3	2014.0	8215	128947375.0	128933450.0	0.965805430316...
4	2015.0	4208	84824900.0	84824900.0	0.974030620725...
5	2016.0	37340	549214275.0	549068925.0	0.976058524698...
6	2017.0	32177	486738000.0	486738000.0	0.972318012976...
7	2018.0	45675	738165450.0	738164450.0	0.986720897056...
8	2019.0	47191	786856600.0	786856600.0	0.988343015741...



04

Anomaly Cohort

Anomaly Cohort Identification - SQL BigQuery

Anomaly Cohort Identification

Cohort dengan TKB30 paling menyimpang dari portfolio average

Anomaly Cohort – Bad Cohort (2012)

TKB30

96.25%

Portfolio OS

\$2.35M

Total Loans

112

Deviation

-1.18%

OS DPD>30

\$88,025

Avg Int Rate

18.36%

Early-stage 2012: underwriting belum matang, 100% tenor 60 bulan, interest rate tinggi

Anomaly Cohort – Good Cohort (2019)

TKB30

98.83%

Portfolio OS

\$786.86M

Total Loans

47,191

Deviation

+1.40%

OS DPD>30

\$9.17M

Avg Int Rate

12.93%

2019: cohort terbaru & terbesar (\$787M OS), underwriting matang, interest rate tidak terlalu tinggi, tenor variatif

Anomaly Cohort Identification

Cohort dengan TKB30 paling menyimpang dari portfolio average

Anomaly Cohort – Bad Cohort (2014)

TKB30

96.58%

Portfolio OS

\$128.95M

Total Loans

8,125

Deviation

-0.85%

OS DPD>30

\$4.41M

Avg Int Rate

13.73%

2014: underwriting belum matang 100% tenor 60 bulan, interest rate tinggi

Anomaly Cohort – Good Cohort (2018)

TKB30

98.67%

Portfolio OS

\$786.86M

Total Loans

45,675

Deviation

+1.24%

OS DPD>30

\$9.80M

Avg Int Rate

12.68%

2018 : underwriting matang, interest rate tidak terlalu tinggi (lebih rendah dibandingkan 2017 & 2019), tenor variatif



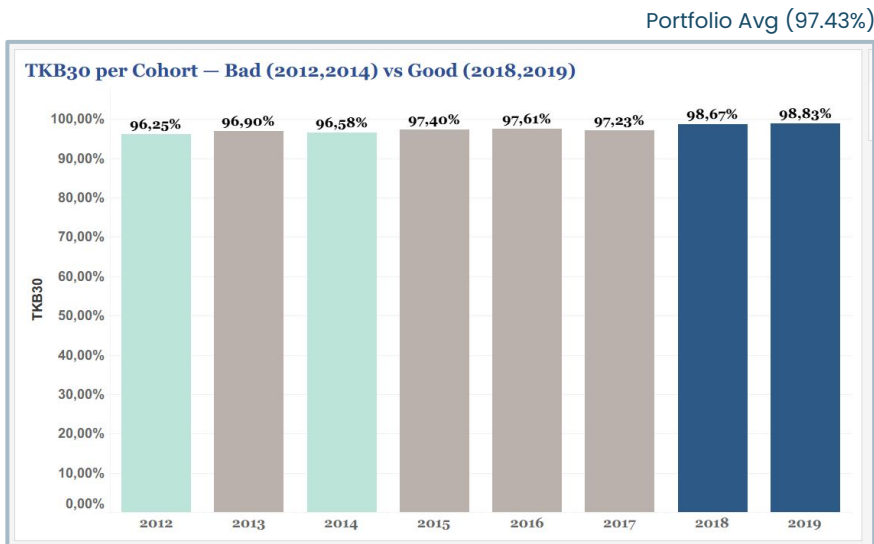
05

Deep Dive : Cohort Analysis

Cohort Analysis – SQL BigQuery

TKB30 per Cohort

Kualitas pembayaran meningkat dari 2012 ke 2019



Terdapat **gap performa ~2% antara Bad Cohort (2012, 2014) dan Good Cohort (2018, 2019)** angka yang **sangat signifikan di dunia credit risk**.

Bad Cohort: 2012 (96.25%) dan 2014 (96.58%) menjadi periode dengan TKB30 terendah, indikasi kualitas underwriting awal yang masih lemah.

Good Cohort: 2018 (98.67%) dan 2019 (98.83%) menunjukkan **peningkatan kualitas portfolio yang konsisten**, hasil dari maturity proses kredit dari waktu ke waktu. Tren TKB30 naik konsisten dari 2012 ke 2019 — mengindikasikan proses underwriting yang semakin matang

Insight ini **memvalidasi business problem** bahwa cohort 2012–2014 menjadi *root of poor repayment performance*, dan menjadi baseline untuk target peningkatan TKB30 sebesar +2–3% di periode berikutnya.

2012-Anomaly (Lowest)

96.25%

Terendah dalam portfolio
-1.18% dari rata-rata

Portfolio Average

97.43%

Benchmark kualitas keseluruhan
cohort

2019-Anomaly (Highest)

98.83%

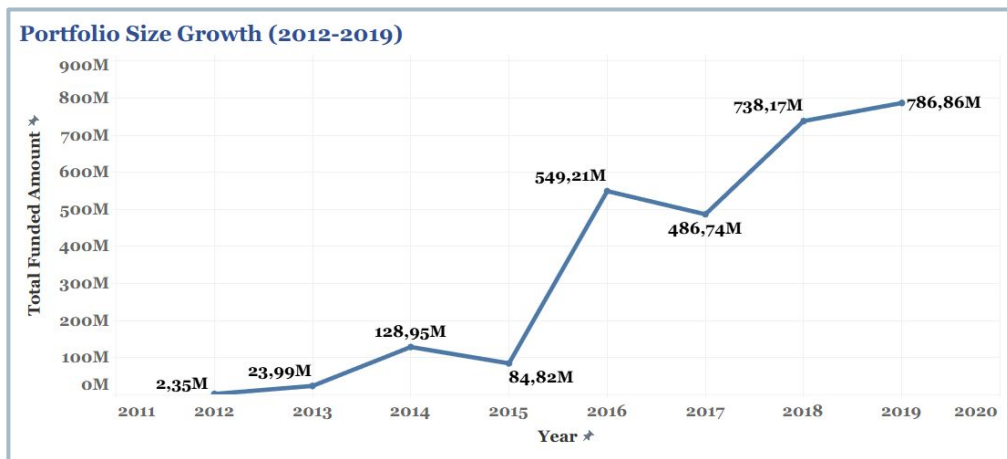
Tertinggi dalam portfolio
+1.40% dari rata-rata

Tren Keseluruhan

+2.58%

Kenaikan TKB30 dari 2012 ke 2019

Portfolio Size Growth

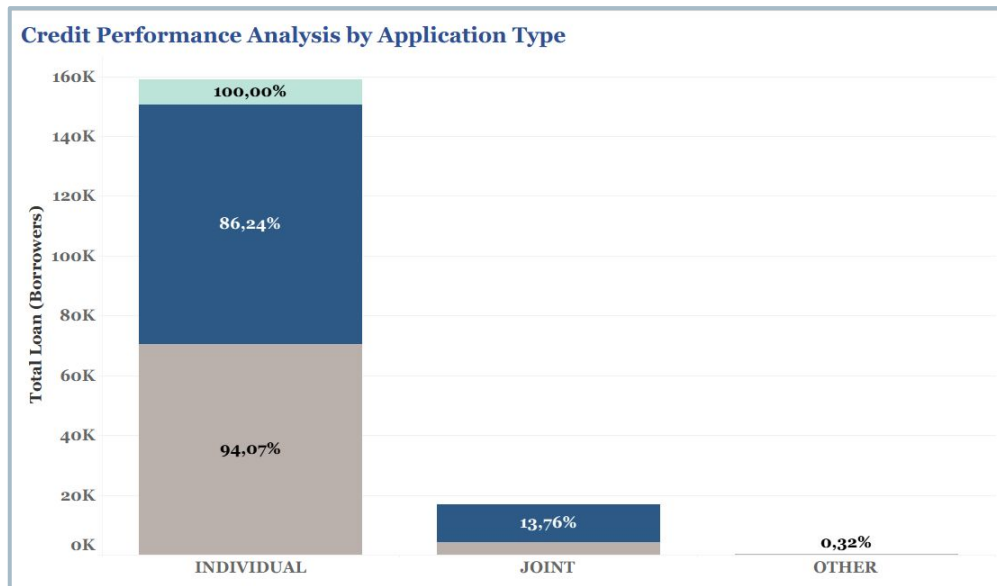


Portfolio RevoFin mengalami **pertumbuhan ekstrem >300x dari \$2.35M (2012) ke \$786.86M (2019)**, menunjukkan ekspansi bisnis yang sangat agresif dalam periode 7 tahun.

Lonjakan signifikan terjadi di 2016 (~\$549M) setelah dip di 2015 (~\$84M), dan kembali tumbuh stabil di 2018–2019. Pola ini mengindikasikan **fase scaling-up yang cepat** setelah masa stabilisasi awal.

Namun **growth bukanlah penyebab langsung penurunan TKB30**. Justru sebaliknya, TKB30 cohort terbaru lebih baik. Artinya, ekspansi ini **diiringi dengan perbaikan standar underwriting**, sehingga risiko default tidak ikut membesar secara proporsional.

Credit Performance by Application Type



Distribusi loan didominasi tipe **INDIVIDUAL (>90% dari total borrowers)**, menjadi baseline market RevoFin.

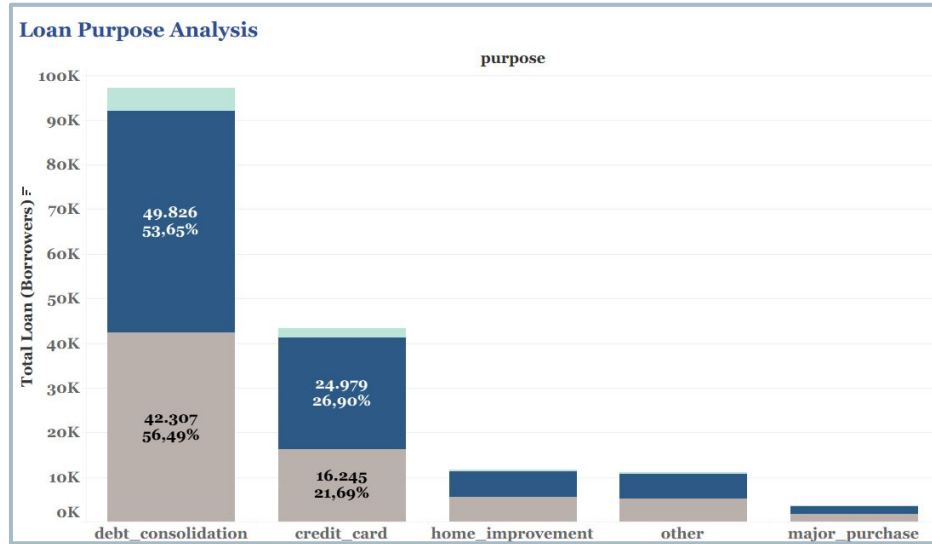
Insight kunci: proporsi **Joint Application di Good Cohort (13.76%) hampir 2.5x lebih besar dibanding Bad Cohort (5.61%)**. Joint loan = dual income → kapasitas repayment lebih kuat → default risk lebih rendah.

Good cohorts have a slightly higher share of joint applications, which typically involve dual income and stronger repayment capacity.

Strategi: **encouraging joint applications dapat menjadi risk mitigation strategy**, misal lewat rate discount atau higher credit limit untuk joint borrowers.

Bad Cohort (2012+2019) Good Cohort (2012+2019) Others

Loan Purpose Analysis



Top kategori loan purpose: **Debt Consolidation (paling dominan)**, **Credit Card**, baru **Home Improvement**, **Other**, **Major Purchase**.

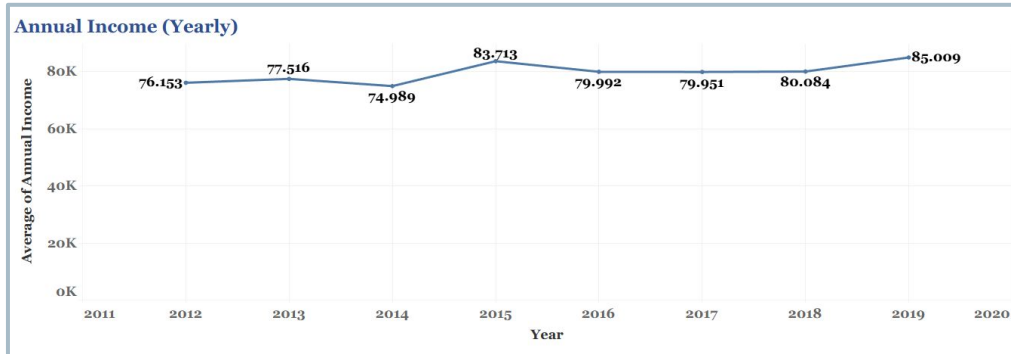
Bad Cohort lebih banyak mengambil loan untuk **Debt Consolidation (~59.85%)** dibanding Good Cohort (~53.65%). Sebaliknya untuk Credit Card, Bad (~24.5%) sedikit lebih rendah dari Good (~26.9%).

Interpretasi: Debt Consolidation = **signal borrower yang sudah punya beban utang sebelumnya**. Biasanya borrower kategori ini sedang refinancing atau menutup utang lama → **kondisi finansial sudah dalam tekanan** → **higher prior debt** → **higher risk** → **default** ↑.

Insight: Bad cohort mengandung lebih banyak borrower dengan masalah utang sebelumnya.

Bad Cohort (2012+2019) Good Cohort (2012+2019) Others

Annual Income (Yearly Trend)



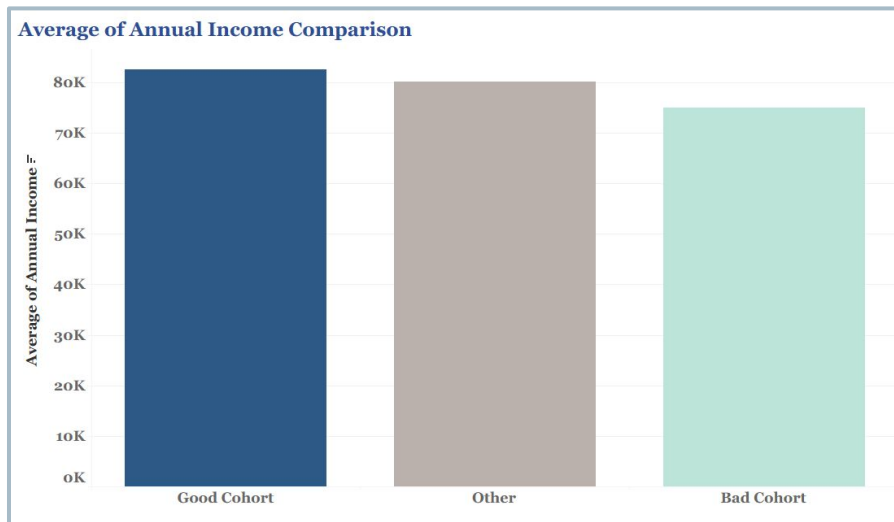
Trend annual income borrower secara umum **meningkat seiring waktu:**

- 2012–2014: ~\$74K – \$77K
- 2015: lonjakan ke ~\$83K
- 2016–2018: stabil di ~\$80K
- 2019: naik kembali ke ~\$85K

Pola ini menunjukkan **kualitas borrower dari sisi kapasitas ekonomi membaik dari waktu ke waktu**, sejalan dengan perbaikan TKB30 di cohort terbaru.

Income yang lebih tinggi di periode terakhir mengindikasikan bahwa **RevoFin berhasil menarik segmen borrower yang lebih solid secara finansial.**

Annual Income – Good vs Bad Cohort



Bad Cohort ≈ \$75K | Good Cohort ≈ \$82–83K

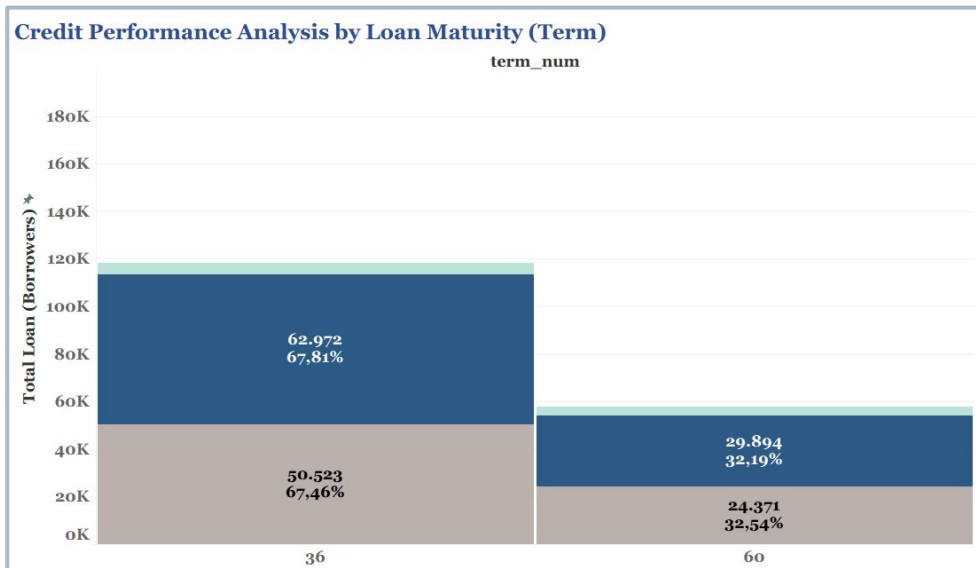
Gap sekitar **\$7K – \$8K (~10% lebih rendah di Bad Cohort)**. Borrowers in bad cohorts exhibit slightly lower income levels (~10% lower), indicating **weaker repayment capacity**.

Namun **gap ini tidak cukup besar untuk menjadi penjelasan tunggal** atas perbedaan TKB30 antar cohort. Income hanya berperan sebagai **secondary driver**.

Implikasi: income perlu dikombinasikan dengan **income-to-installment ratio** sebagai threshold approval, bukan sebagai filter tunggal di underwriting.

■ Bad Cohort (2012+2019) ■ Good Cohort (2012+2019) ■ Others

Credit Performance by Loan Maturity (Term)



RevoFin menawarkan 2 tenor utama: **36 bulan dan 60 bulan**.

Distribusi tenor 36 bulan: Good 67.81% | Bad 56.41%

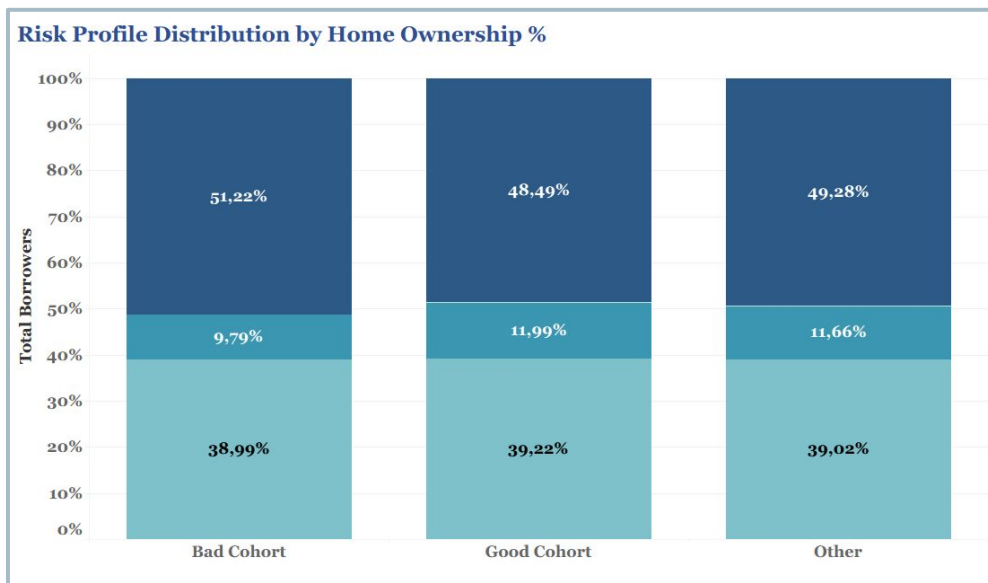
Distribusi tenor 60 bulan: Good 32.19% | **Bad 43.59%**

Bad Cohort **lebih banyak mengambil tenor panjang (60 bulan) – 43.6% vs hanya 32.2% di Good Cohort**. Sebaliknya, Good Cohort lebih dominan di tenor pendek 36 bulan.

Insight: tenor panjang = **exposure duration lebih besar** → **repayment burden bertambah** → **borrower fatigue** → **default risk meningkat**. Loan tenor menjadi **PRIMARY driver** penurunan TKB30 di Bad Cohort.

Bad Cohort (2012+2019) Good Cohort (2012+2019) Others

Risk Profile by Home Ownership



Komposisi home ownership **hampir identik antara Bad dan Good Cohort:**

Bad Cohort: Mortgage 51.22% | Rent 38.99% | Own 9.79%

Good Cohort: Mortgage 48.49% | Rent 39.22% | Own 11.99%

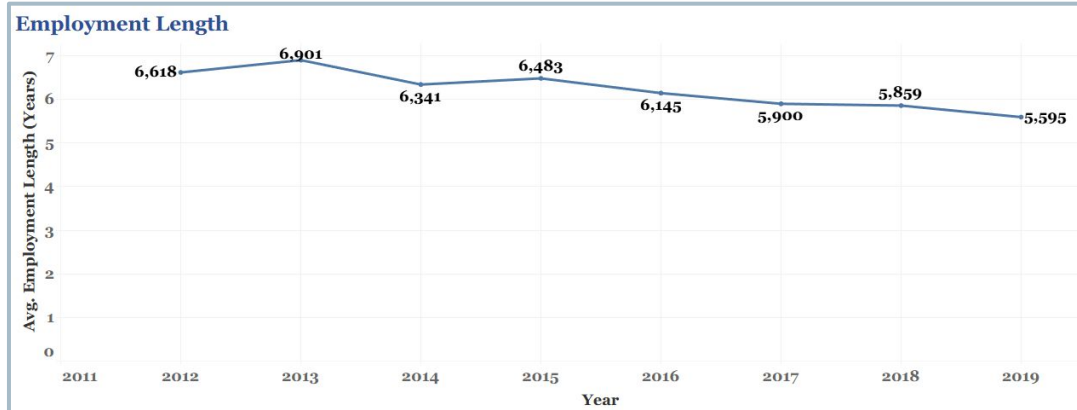
Selisih antar cohort hanya berkisar **1-3 percentage points** (terlalu kecil untuk dianggap sebagai pembeda performa).

Insight: **Home ownership TIDAK menjadi pembeda signifikan** antara Bad vs Good cohort, sehingga **bukan root cause utama rendahnya TKB30.**

Rekomendasi: deprioritaskan home ownership sebagai variabel screening utama; pertahankan hanya sebagai fitur minor di model scoring.

RENT MORTGAGE OTHER OWN

Employment Length (Yearly Trend)



Trend employment length **menurun dari waktu ke waktu**:

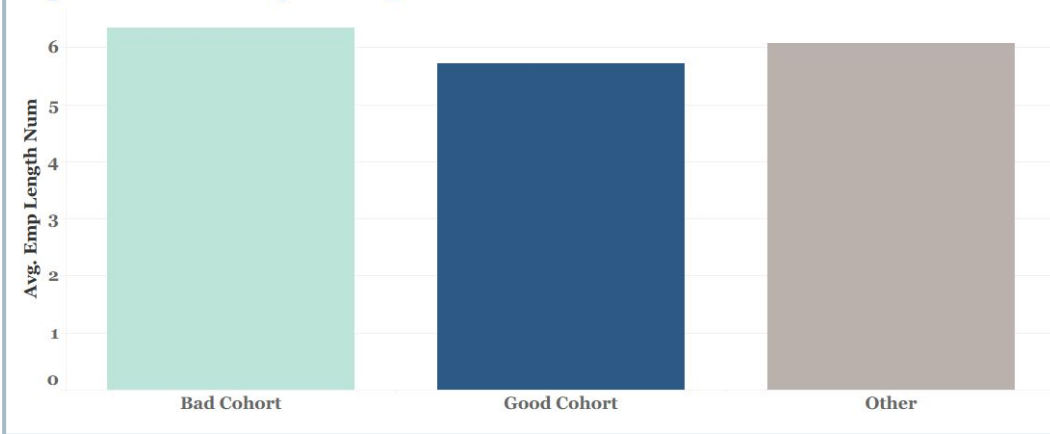
- 2012–2014: ~6.6 – 6.9 tahun
- 2015–2017: ~5.9 – 6.5 tahun
- 2018–2019: ~5.6 – 5.9 tahun

Borrower RevoFin makin **“junior”** (tenure kerja lebih pendek). Pola ini menarik karena **biasanya tenure lebih lama → lebih stabil → performa lebih baik**. Tapi di data ini, justru **borrower yang lebih muda (cohort terbaru) memiliki TKB30 yang lebih bagus**.

Kemungkinan: market shift menuju younger borrowers yang lebih ter-screen / lebih disiplin secara digital.

Impact of Job Seniority on TKB30

Impact of Job Seniority on TKB30: Good vs. Bad Cohorts



Counterintuitive finding:

- Bad Cohort (2012, 2014) → employment length lebih **tinggi**
- Good Cohort (2018, 2019) → employment length lebih **rendah**

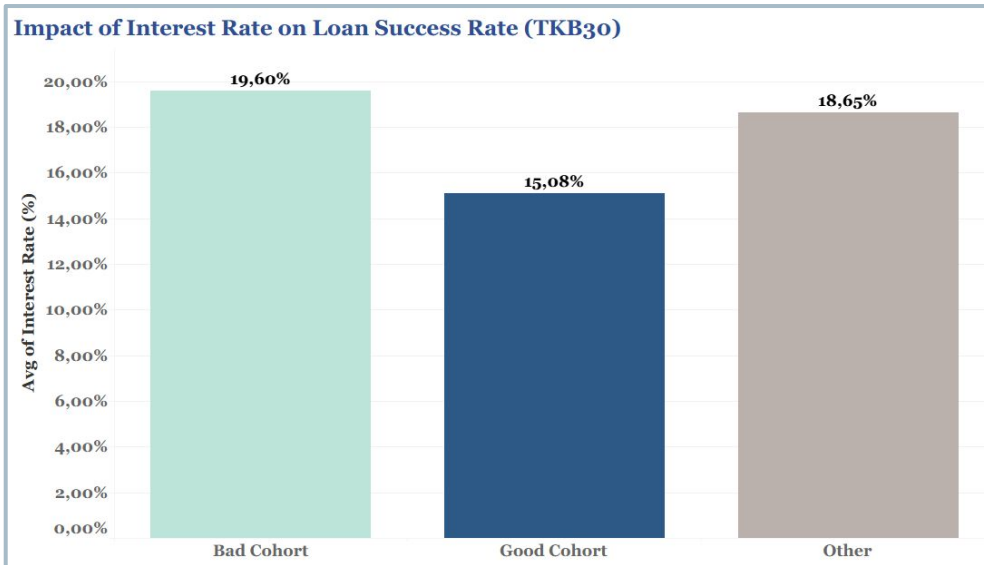
Realita ≠ ekspektasi. Employment length TIDAK menjelaskan kenapa Bad Cohort lebih jelek, bahkan polanya berlawanan.

Kemungkinan interpretasi:

1. Borrower dengan tenure lebih lama justru **punya beban debt/obligation yang lebih banyak**
2. Younger borrowers di cohort terbaru sudah melalui **credit scoring yang lebih ketat**

Rekomendasi: jangan gunakan employment length sebagai proxy tunggal untuk financial stability.

Impact of Interest Rate on TKB30



Bad Cohort: 19.60% | Good Cohort: 15.08% | Other: 18.65%

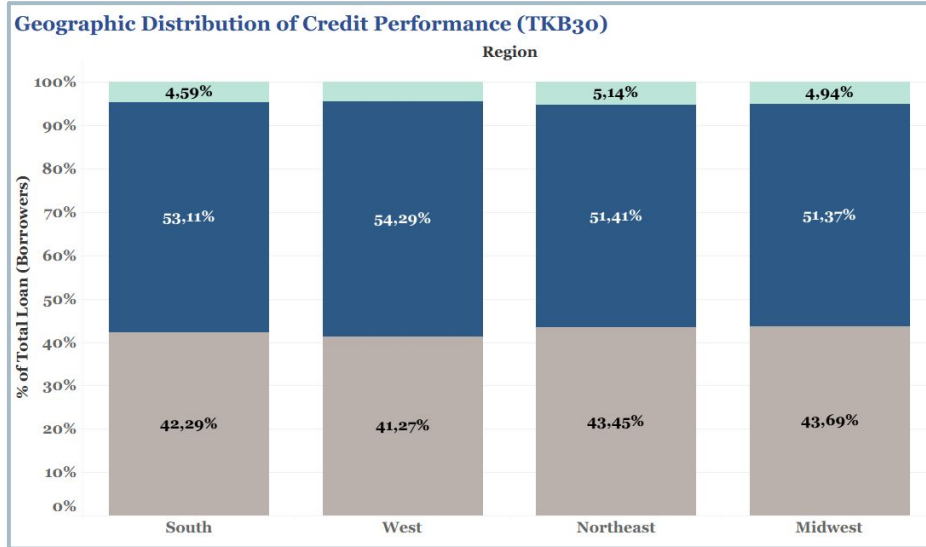
Gap interest rate ~+4.5 percentage points lebih tinggi di Bad Cohort, ini **SIGNIFIKAN** dibanding income gap (~10%) atau employment length (kebalik).

*Higher interest rates in bad cohorts suggest that riskier borrowers were charged more, but instead of compensating for risk, this may have **increased repayment burden and worsened performance.***

Rantai sebab: Interest ↑ → cicilan bulanan ↑ → repayment stress ↑ → default ↑ → TKB30 ↓

Insight: Interest Rate adalah **PRIMARY DRIVER** utama default. Solusi: dynamic interest rate cap berbasis credit scoring, dengan pricing tier yang ketat agar high-risk borrower tidak otomatis diberi rate tinggi tanpa mitigasi lain.

Geographic Distribution of Credit Performance



Distribusi performa kredit di empat region utama:

- South: 53.11% / 42.29%
- West: 54.29% / 41.27%
- Northeast: 51.41% / 43.45%
- Midwest: 51.37% / 43.69%

Selisih antar region **sangat kecil (<3 percentage points)** — distribusi performa pinjaman **relatif merata secara geografis**.

Insight: **Region BUKAN diferensiator utama TKB30**. Masalah default tersebar merata di seluruh region, sehingga **strategi mitigasi tidak perlu region-specific** — fokus tetap pada loan structure (interest rate & tenor) sebagai primary driver.

Rekomendasi: terapkan kebijakan underwriting yang seragam secara nasional, dengan monitoring ekstra di kantong-kantong default tertinggi.

■ Bad Cohort (2012+2019) ■ Good Cohort (2012+2019) ■ Others

Deep Dive: Bad vs Good Cohort Profiling

Perbandingan profil — Bad Cohort (2012, 2014) vs. Good Cohort (2018, 2019)

Dimensi Profil	Bad Cohort (2012, 2014)	Good Cohort (2018, 2019)	Signifikansi
Home Ownership	MORT: 51.22% RENT: 38.99% OWN: 9.79%	MORT: 48.49% RENT: 39.22% OWN: 11.99%	Tidak Signifikan
Avg Employment Length	6.6 – 6.9 tahun	5.6 – 5.9 tahun	Tidak Signifikan
Annual Income (Avg)	~\$75K	~\$82 – \$83K	Sedang
Top Loan Purpose	Debt Consol: 59.85% CC: 24.49%	Debt Consol: 53.65% CC: 26.90%	Sedang
Loan Type — Joint	5.61%	13.76%	Sedang
Avg Interest Rate	19.60% ← KRITIS	15.08%	SANGAT KRITIS
Term 60 Months	43.59% ← KRITIS	32.19%	SANGAT KRITIS



06

Insight & Recommendation

OBIPR Framework

OBIPR Framework – Insight & Rekomendasi

Observation–Business Impact–Isolation–Prioritization–Recommendation

Observation (Temuan) / Category	Business Impact (Dampak Bisnis)	Isolation (Area Fokus)	Prioritization (Prioritas)	Recommendation (Rekomendasi)
PRIMARY Interest Rate Gap	Bad cohort memiliki bunga rata-rata 19.6% vs good cohort 15.08%, terdapat selisih +4.5% langsung meningkatkan cicilan bulanan dan beban repayment borrower → TKB30 menurun signifikan	Bandingkan distribusi int_rate antara bad cohort (2012, 2014) vs good cohort (2018, 2019); analisis korelasi interest rate terhadap default rate per cohort	High – Primary driver	Tambahkan layer screening khusus untuk debt consolidation: analisis existing debt load, debt-to-income ratio, dan payment history sebelum approval; pertimbangkan rate premium untuk segmen ini
PRIMARY Loan Tenor Mismatch	Bad cohort memiliki proporsi tenor 60 bulan 43.6% vs good cohort hanya 32.2%. Kombinasi high interest + long tenor menciptakan <i>total repayment burden</i> yang besar → borrower fatigue → default naik	Analisis distribusi term_num (36 vs 60 bulan) per cohort; bandingkan completion rate dan default rate antar tenor	High – Primary driver	Batasi akses tenor 60 bulan hanya untuk borrower dengan credit profile kuat; tambahkan syarat income minimum untuk loan jangka panjang; kombinasikan dengan rate cap untuk mengurangi dual burden
SECONDARY Income Gap	Bad cohort memiliki annual income rata-rata ~75K vs good cohort ~82–83K, terdapat selisih ~10% menunjukkan kapasitas repayment yang lebih lemah, meski tidak cukup besar untuk menjadi penyebab tunggal	Analisis distribusi annual_inc per cohort; bandingkan median income antara defaulted vs fully paid borrower	Medium – Secondary driver	Perkuat verifikasi income di underwriting; gunakan income-to-installment ratio sebagai threshold approval; pertimbangkan annual_inc_joint sebagai faktor positif dalam scoring
SECONDARY Debt Consolidation Risk	Bad cohort memiliki proporsi debt consolidation 59.85% vs good cohort 53.65%. Borrower dengan tujuan ini umumnya sudah memiliki beban utang sebelumnya → financial stress lebih tinggi → risiko default meningkat	Analisis conversion rate per loan purpose; bandingkan default rate antara debt consolidation vs credit card vs home improvement	Medium – Risk indicator	Tambahkan layer screening khusus untuk debt consolidation: analisis existing debt load, debt-to-income ratio, dan payment history sebelum approval; pertimbangkan rate premium untuk segmen ini

OBIPR Framework – Insight & Rekomendasi

Observation–Business Impact–Isolation–Prioritization–Recommendation

SECONDARY Loan Type (Joint vs Individual)	Good cohort memiliki proporsi joint loan 13.76% vs bad cohort hanya 5.61%. Joint = dual income → repayment capacity lebih kuat → default risk lebih rendah	Bandingkan default rate antara individual vs joint borrower per cohort; analisis income joint vs individual	Medium — Support factor	Insentifkan joint applications lewat rate discount atau limit yang lebih tinggi; sosialisasi keuntungan joint loan kepada calon borrower; prioritaskan joint dalam approval jika borderline
CONTEXT Portfolio Growth Shock	Portfolio tumbuh ekstrem dari ~2M (2012) ke ~780M (2019) (signifikan pertumbuhan nya >300x). Ekspansi cepat berpotensi melonggarkan standar underwriting → kualitas borrower mix menurun di periode ekspansi	Analisis distribusi loan_status per tahun; bandingkan TKB30 cohort dengan kecepatan pertumbuhan volume pada tahun yang sama	Low — Contextual driver	Terapkan growth-adjusted underwriting standard: saat volume tumbuh cepat, perkuat, bukan longgarkan, standar approval; lakukan cohort monitoring per kuartal untuk deteksi dini penurunan kualitas
CONTEXT Home Ownership (Non-driver)	Komposisi home ownership hampir identik antara bad cohort (mortgage 51.22%, rent 38.99%) dan good cohort (mortgage 48.49%, rent 39.22%), tidak ada perbedaan signifikan	Distribusi home_ownership_clean sudah dianalisis per cohort; hasilnya menunjukkan tidak ada discriminating power	Low — Non-significant	Kurangi prioritaskan home ownership sebagai variabel screening utama; alihkan fokus model ke interest rate, tenor, dan income. Pertahankan sebagai fitur minor dalam model scoring untuk kelengkapan
CONTEXT Employment Length (Counterintuitive)	Bad cohort (2012–2014) memiliki emp_length lebih tinggi (6.6–6.9 tahun) vs good cohort (2018–2019) yang lebih rendah (5.6–5.9 tahun) — berlawanan dengan ekspektasi. Emp_length tidak membedakan performa cohort	Analisis distribusi emp_length_num per cohort dan per loan_status; test korelasi dengan TKB30	Low — Counterintuitive, weak signal	Jangan gunakan employment length sebagai proxy tunggal untuk financial stability; kombinasikan dengan income level dan debt profile untuk penilaian kapasitas yang lebih akurat



Appendix





RevoFin Loan Portfolio Insights: Cohort Risk Analysis Dashboard

Cohort Group

(All)

Issue Year

(All)

Loan Status

(Multiple values)

92,866 loans

Good Cohort (2012-2019)

8,327 loans

Bad Cohort (2012-2019)

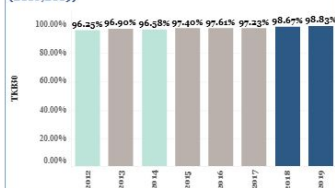
98.11%

TKB30

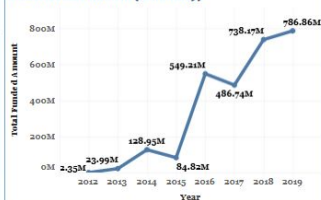
\$2.80B

Portfolio OS

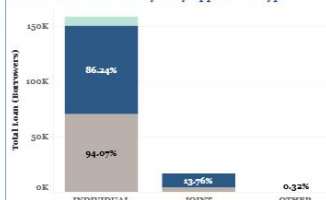
TKB30 per Cohort – Bad (2012,2014) vs Good (2018,2019)



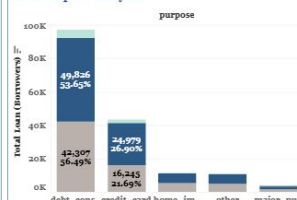
Portfolio Size Growth (2012-2019)



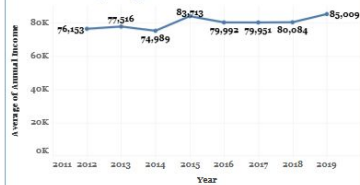
Credit Performance Analysis by Application Type



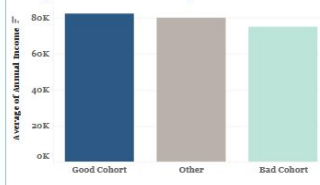
Loan Purpose Analysis



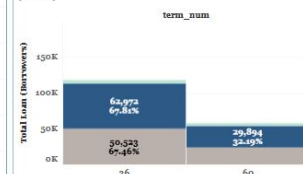
Annual Income (Yearly)



Average of Annual Income Comparison



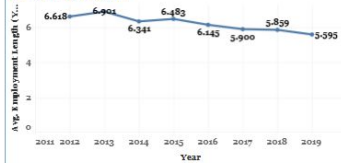
Credit Performance Analysis by Loan Maturity (Term)



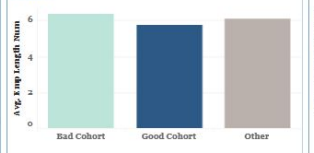
Risk Profile Distribution by Home Ownership %



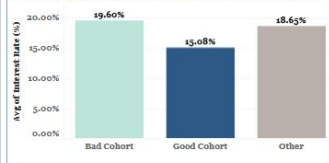
Employment Length



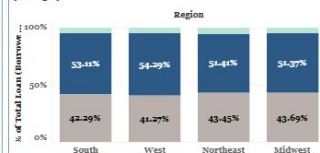
Impact of Job Seniority on TKB30: Good vs. Bad Cohorts



Impact of Interest Rate on Loan Success Rate (TKB30)



Geographic Distribution of Credit Performance (TKB30)



RevoFin Dashboard

[Link Dashboard Tableau Public](#)

Data Cleaning

SQL Bigquery Code Documentation

STEP 1

Clean up the customer_id column on customer and loan data (table). Use the following BigQuery functions to do so

```
array_to_string(regexp_extract_all(customer_id, "[a-zA-Z0-9]"), "")
```

```
SELECT
  ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id,
    "[a-zA-Z0-9]"), "") AS cleaned_customer_id
FROM `nextproject-001.FIP.customer`
```

```
SELECT
  ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id,
    "[a-zA-Z0-9]"), "") AS cleaned_customer_id
FROM `nextproject-001.FIP.loan`
```

STEP 2

Find all loans that are still in the portfolio, defined as loans with status Current, In Grace Period, Late, and Default

```
SELECT *
FROM loan_clean
WHERE loan_status IN (
  'Current',
  'In Grace Period',
  'Late (16-30 days)',
  'Late (31-120 days)',
  'Default'
```

Data Cleaning

Clean table 'customer'

```
-- CLEAN CUSTOMER TABLE
WITH customer_clean AS (
  SELECT
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id, "[a-zA-Z0-9]"), "") AS
    cleaned_customer_id,
    annual_inc,

    -- emp_length jadi numeric
    CAST(REGEXP_EXTRACT(emp_length, r'\d+') AS INT64) AS emp_length_num,

    -- home ownership clean
    CASE
      WHEN home_ownership IN ('ANY', 'NONE', 'OTHER') THEN 'OTHER'
      ELSE home_ownership
    END AS home_ownership_clean

  FROM `nextproject-001.FIP.customer`
)

SELECT *
FROM customer_clean;
```

Clean table 'loan'

```
-- CLEAN LOAN TABLE
WITH loan_clean AS (
  SELECT
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id, "[a-zA-Z0-9]"), "") AS cleaned_customer_id,

    funded_amount,
    loan_status,
    issue_year,
    purpose,
    state,
    int_rate,

    -- type clean
    CASE
      WHEN UPPER(type) IN ('JOINT', 'JOINT APP') THEN 'JOINT'
      WHEN UPPER(type) = 'INDIVIDUAL' THEN 'INDIVIDUAL'
      ELSE 'OTHER'
    END AS type_clean,

    -- term numeric
    CAST(REGEXP_EXTRACT(term, r'\d+') AS INT64) AS term_num

  FROM `nextproject-001.FIP.loan`
)

SELECT *
FROM loan_clean;
```

Data Cleaning (cleaned customer_id)

Clean table 'customer'

Row	cleaned_customer_id	annual_inc	emp_length_num	home_ownership_clean
1	bk0Hjxc6x99xc8f8Cx80xb5x16hxa4xcced6x01yxa7xdaySx92xe7xc4xe7x17x86xd20Q	28000.0	null	OTHER
2	bxe8txc7xfxc8Bxb91xbdmxe8xb0x1fuxd0qx0fx17jx03xfxc95xeexc0x10xc69	173005.0	null	OTHER
3	bx97Dx95xc6xe5x10x98x91xcfxd4x9bxc3xccf4x9bX03x9ax1dx90x16xc3Mxe4xb2xaa	98000.0	null	OTHER

Results per page: 50 1 - 50 of 270299

Row	cleaned_customer_id	annual_inc	emp_length_num	home_ownership_clean
4	bx1cbxa4Lxd9xc8Sxa2x17xb8xb3x9d31x9cxbxaeaTlGxae88x9S8e8lx9bxd8	47000.0	null	OTHER
5	bx3d3f9xe6fbxcfxe9xa4xa9N...	15576.0	null	OTHER
6	bxecUWx07Zxa4wxdcxb3x9bxc1Xxd82x8ex00x1a5xcd0x12x0bxd1x82x15Px7x92N	28000.0	null	OTHER
7	bx13xad8cxf4xbex8fxc12x85xaxcd93XFxfxbda5x8c0Dxb	15300.0	null	OTHER

Results per page: 50 1 - 50 of 270299

Clean table 'loan'

Row	cleaned_customer_id	funded_amount	loan_status	issue_year	purpose	state
1	bx02cx5xd0xec1exbxd07xb5x...	40000.0	Current	2017.0	renewable_energy	ID
2	bXxa7xa9x13yx8cxdax10aXx87...	25000.0	Fully Paid	2013.0	debt_consolidation	OK
3	bKxaax15xddwa88xx8bxfex026x98xe5fx8cx87xf6xbdx1fx03.jxd6xdcl.x8fx8abxexd1	6000.0	Fully Paid	2012.0	car	MN
4	bx06xe9x1d4x80ex19x069xc1xd6wx8cx13x07x9ex95x81xb0xf9x88xc3x1bxe2xccx95axd	5525.0	Fully Paid	2012.0	debt_consolidation	OR

Results per page: 50 1 - 50 of 270299

Row	cleaned_customer_id	funded_amount	loan_status	issue_year	purpose	state
5	bnx1fxfxud7yxb7xc4xcxb1eMxdxb5x99xdaxfarUExc2xaejxd8Wx1bxd3Cxb	7500.0	Fully Paid	2013.0	car	GA
6	bx1exf3xcbpWnxb1x84xd8xd5xb4dxc95xa1x87qxa4xc6JEx0cx97fxe4xe9x85X	8000.0	Fully Paid	2012.0	credit_card	VA
7	bx11xa7xfce08Nxc4xb4xecxc3xb4x83.jxebx2nDxb2xb0xf3xd7xb6xc7xc1xdxbfbxb6x9a	9750.0	Fully Paid	2013.0	home_improvement	TN

Results per page: 50 1 - 50 of 270299

Executive Summary

SQL Bigquery Code Documentation

STEP 3

Based on the loan's funded amount and cleaned customer_id, calculate the number of customers in the portfolio, total funded amount (call it outstanding amount/OS), and current ENR of the portfolio. Note: Because we don't directly observe DPD, you can assume that all Default loan's OS is not included in the ENR.

STEP 4

Calculate the TKB30 of the portfolio, defined as

$$\text{TKB30} = 1 - (\text{OS of loan with DPD} > 30 \text{ days}) / (\text{Portfolio OS})$$

Result Query:

Row	total_customers	portfolio_os	enr	tkb30
1	176087	2801078850.0	2800883575.0	0.981063635534...

Executive Summary (3 & 4)

```
-- EXECUTIVE SUMMARY NUMBER 3 & 4 --
WITH loan_clean AS (
  SELECT
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id, "[a-zA-Z0-9]"), "") AS
cleaned_customer_id,
    funded_amount,
    loan_status
  FROM `nextproject-001.FIP.loan`
),

portfolio AS (
  SELECT *
  FROM loan_clean
  WHERE loan_status IN (
    'Current',
    'In Grace Period',
    'Late (16-30 days)',
    'Late (31-120 days)',
    'Default'
  )
)
```

```
SELECT
  COUNT(DISTINCT cleaned_customer_id) AS total_customers,

  -- OS
  SUM(funded_amount) AS portfolio_os,

  -- ENR
  SUM(CASE
    WHEN loan_status != 'Default'
    THEN funded_amount
    ELSE 0
  END) AS enr,

  -- TKB30
  1 - (
    SUM(CASE
      WHEN loan_status IN ('Late (31-120 days)', 'Default')
      THEN funded_amount
      ELSE 0
    END)
    /
    SUM(funded_amount)
  ) AS tkb30

FROM portfolio;
```

Executive Summary

SQL Bigquery Code Documentation

STEP 5

Calculate OS, ENR, and TKB30 of each loan cohort. You can define cohorts on a monthly or yearly basis

STEP 6

From your findings in (5), is there any particular cohort with high or low TKB30?

Result Query:

Row	cohort_year	total_customers	os	enr	tkb30
1	2012.0	112	2346175.0	2346175.0	0.962481485822...
2	2013.0	1169	23986075.0	23951075.0	0.968954904043...
3	2014.0	8215	128947375.0	128933450.0	0.965805430316...
4	2015.0	4208	84824900.0	84824900.0	0.974030620725...
5	2016.0	37340	549214275.0	549068925.0	0.976058524698...
6	2017.0	32177	486738000.0	486738000.0	0.972318012976...
7	2018.0	45675	738165450.0	738164450.0	0.986720897056...
8	2019.0	47191	786856600.0	786856600.0	0.988343015741...

Executive Summary (5)

```
-- EXECUTIVE SUMMARY NUMBER 5 --

WITH loan_clean AS (
  SELECT
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id, "[a-zA-Z0-9]"), "") AS
cleaned_customer_id,
    funded_amount,
    loan_status,
    issue_year
  FROM `nextproject-001.FIP.loan`
),

portfolio AS (
  SELECT *
  FROM loan_clean
  WHERE loan_status IN (
    'Current',
    'In Grace Period',
    'Late (16-30 days)',
    'Late (31-120 days)',
    'Default'
  )
)
```

```
SELECT
  issue_year AS cohort_year,
  COUNT(DISTINCT cleaned_customer_id) AS total_customers,
  SUM(funded_amount) AS os,
  SUM(CASE
    WHEN loan_status != 'Default' THEN funded_amount
    ELSE 0
  END) AS enr,
  1 - SAFE_DIVIDE(
    SUM(CASE
      WHEN loan_status IN ('Late (31-120 days)', 'Default')
      THEN funded_amount
      ELSE 0
    END),
    SUM(funded_amount)
  ) AS tkb30
FROM portfolio
GROUP BY issue_year
ORDER BY issue_year;
```



Thanks!

Any questions?

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+62 818-970-500

[Link Google Big Query](#)

[Link Google Colab
\(Documentation\)](#)

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